

Program Update Day Wednesday 22 January 2020 (Bolognalaan 101, room 0.204)

<i>Time</i>	<i>UU Supervisor</i>	
09.00 - 09.15	Welcome	
09.15 - 09.30	Carlijn van de Pas	Terburg
09.30 - 09.45	Hanna Sundermeier	Beckers
09.45 - 10.00	Kaja Sramova	Meijer
10.00 - 10.10	<i>Poster pitch session 1</i>	
10.10 - 10.30	<i>Break</i>	
10.30 - 10.45	Tiemen Wagenvoort	Benjamins
10.45 - 11.00	Onne Sijbrandij	Donker
11.00 - 11.15	Monica Rapisarda Botero	Donker
11.15 - 11.30	Jonas kamps	Bijlsma
11.30 - 11.40	<i>Poster pitch session 2</i>	
11.40 - 12.00	<i>Break</i>	
12.00 - 12.15	Anouk Ticheler	Dennis
12.15 - 12.30	Anna-Sophie Laan	Meijer
12.30 - 12.45	Julia van der Donk	Meijer
12.45 - 13.00	Pim Verhagen	Overvliet
13.00 - 13.10	<i>Poster pitch session 3</i>	
13.10 - 14.00	<i>Lunch</i>	
14.00 - 14.15	Claire Borst	Romeijn
14.15 - 14.30	Helen Visser	Romeijn
14.30 - 14.45	Bianca Beerepoot	Romeijn
14.45 - 15.30	<i>Poster session</i>	
15.30 - 17.00	<i>Drinks</i>	

Poster pitch Session1	
<i>Name</i>	<i>UU Supervisor</i>
Molly Adams	Baas
Josefine Funnekotter	Beckers
Iga Jarosz	de Kloe
Julia Meijer	Donker
Nina Geldof	Donker
Nicole Plungjan	Donker
Poster pitch Session2	
Prithvi Dheeraj Cheraku Mulanti	Keizer
Lisa Verheul	Meessen
Job Davidse	Meijer
Lars Dijkstra	Mulder
Poster pitch Session 3	
Vani Pillay	Overvliet
Elina Heeres	Romeijn
Perseus Zhao	Naber
Ditte van Haalen	Terburg
Anais Thijssen	Baas

Abstracts oral presentations

Name Student: Carlijn van de Pas

Size of project: 12,5ECTS

Title project: Making the life of a researcher easier; A User-Experience Research into dashboards

Abstract: Metrixlab is a global market research company and is expert in brand engagement, product innovation and customer value. Many researchers work with dashboards, which provide their clients with real time data about products, websites, campaign effectiveness, etcetera. Experience has shown that the researchers at the Metrixlab office don't like working in the dashboard environment, because in their opinion it is not user friendly. Therefore, I will conduct a user-experience research into these dashboards during my internship. By conducting multiple interviews in the usability lab, I will get more insight in the user stories and experience of the employees. This will enable me to provide Metrixlab with recommendations on how to improve their dashboard environment. Every two weeks the Metrixlab development team will implement some of the recommendations, which will give me the opportunity to immediately test the changes in the next interviews. Next to the usability interviews I have made an online survey which the employees can complete on every page of the dashboard environment. This will provide insight into whether the user friendliness has improved after implementing some of my recommendations.

Location project (name company or University): Metrixlab

External supervisor: Sanne Handgraaf

UU Supervisor: David Terburg

Name Student: Hanna C. Sundermeier

Size of project: 12.5 ECTS

Title project: Rethinking work processes at AkzoNobel

Abstract: Procurement processes in AkzoNobel have recently changed, now people need to learn about the new process, get used to it and be prepared for when things do not run smoothly. As part of the change management team I created a game/e-learning to teach 500 people across the globe about this new procurement process.

Location project (name company or University):AkzoNobel

External supervisor: Anne Wilhelmus

UU Supervisor: Gabriel Beckers

Student name: Kaja Sramova

Size of project: 12.5 ECTS

Title project: Implementation of an international community-based mental health care project

Abstract: RECOVER-E is a large-scale multinational project, coordinated by mental health institutes in the Netherlands and Spain. The project strives to implement community-based mental health care for people with severe and enduring mental health problems in five countries in Europe (Bulgaria, Croatia, Montenegro, North Macedonia, and Romania). The aim of the individual internship project is to interview team leaders from these five implementation sites on their experience with the RECOVER-E project, in order to learn the strengths and weaknesses of the project. Based on the findings of the qualitative analysis, an advisory report will present what measures the coordinating

side should take to support the implementation in the five sites.

Location project (name company or University): Trimbos Instituut, Utrecht

External supervisor: Bethany Hipple Walters

UU Supervisor: Larissa Meijer

Name Student: Tiemen Wagenvoort

Size of project: 7.5 ECTS

Title project: Kijken, kijken, niet kopen? (To look, to look, but not to buy?) Using nudging and mobile eye tracking to promote and investigate sustainable food choices in a canteen

Abstract: In this internship I did a preliminary research on the Future Food Lab. With this project, Green Office Utrecht University, Future Food and Sodexo try to make Utrecht University's canteens more sustainable, but sales of the Future Food Lab are sub optimal. To gain more insight in the situation, I observed walking patterns of customers and conducted pilot measurements with a mobile eye tracker. I will use the results of this internship in my thesis, in which nudging will be used to promote products of the Future Food Lab. Additionally, I will use mobile eye tracking and questionnaires to gain more knowledge on decision processes and perception regarding sustainable food.

Location project (name company or University): Utrecht University

External supervisor: -

UU Supervisor: Jeroen Benjamins

Name Student: Onne Sijbrandij

Size of project: 12,5 ECTS

Title project: The influence of travel time perception on the valuation of Park and Ride facilities

Abstract: The municipality of Utrecht uses a traffic model to model the flow of traffic through the city and to predict future traffic situations. The city's ambition of halting the growth of traffic calls for other alternatives to reach the inner city. P+R facilities are used as incentives to capture car traffic at the city outskirts and transfer this traffic from the outskirts to the centre by public transport instead. The traffic model has not adopted the psychological effect of perceived time multipliers during such transfers which result in time penalties on perceived travel time and may lead to other choices of transport. In this research a literature study is performed to find aspects of travel that are suitable to map a visitor's behaviour of P+R facilities. The aspects of travel that are found are subject to certain perceived time multipliers which cause the user of a P+R facility to perceive time here as longer than in-vehicle time. Furthermore, a survey is executed to find to what extent the aspects of travel from the literature study are of influence at the Westraven P+R. These outcomes are used to give a quality score to each individual aspect of travel at the facility, as well as an overall quality score for the P+R. Apart from this quality score, a framework is also created which can be used as a tool to calculate resistance parameters of this and other P+R facilities.

Location project (name company or University): Gemeente Utrecht (Municipality of Utrecht)

External supervisor: Dick Terlouw

UU Supervisor: Stella Donker

Name Student:: Monica Rapisarda Botero

Size of project: 12.5 ECTS

Title of project: Human factors in medical manufacturing

Abstract: I undertook my curricular internship at Emergo by UL's Human Factors Research and Development office in Utrecht. UL is a global consultancy in the safety industry with offices in 46 countries. The Human Factors division provides human factors guidance and services to medical device manufacturers who are seeking approval by regulatory bodies

such as the American FDA. As an intern at a consultancy, rather than at a product company or institute, I participated in multiple projects concurrently rather than focusing on one single study from inception to end. Some of my tasks over the course of the three months at the office were, for example, assisting in the creation of usability testing materials, acting as a data analyst during usability testing, carrying out research on reported adverse events for a medical device, preparing protocols for testing and ghost writing reports to be submitted for regulatory approval. My work spanned multiple device types: from epinephrine autoinjectors, to laboratory meters, to industrial kitchen appliances. In turn, I gained a broad understanding of the varied field of human factors and its processes and obtained experience in the consultancy sector.

Location project: UL

External supervisor: Veerle Migchelbrink

UU supervisor: Stella Donker

Name Student: Jonas Kamps

Size of project: 12.5 ECTS

Title project: MRI patient coaching

Abstract: I am working on four tasks in the MR patient coaching project at Philips in Eindhoven. The project seeks to improve patient experience and engagement around and during diagnostic MRI scans. I have assembled a detailed literature overview that captures the current academic state of the art in MRI preparation and coaching, and published it as a technical note within Philips. Further, I wrote a short advice report on an MRI patient coaching application currently in development within the project, discussing its content and UI. My advice was introduced in a development meeting. I have thus learned about literature research with a project focus and about the application of literature in the prototyping process of an app. Until the end of February, I will join one of my supervisors as she tests new patient guidance and coaching products for MRI scans. Participating in these studies, I will see how more mature prototypes are tested. Lastly, I will help write a publication about a similar feasibility study from last year, the results of which have not yet been analysed and published. This will deepen my knowledge about sharing of results within a company and beyond. I am learning a great amount about major procedures in corporate research.

Location project (name company or University): Philips Research Eindhoven

External supervisor: Sanne Naats

UU Supervisor: Martijn Bijlsma

Name Student: Anouk Ticheler

Size of project: 12,5 ECTS

Title project: Evaluation of the 'Personal Approach Pilot' and usability research of the smart thermostat 'Toon'

Abstract: Utrecht Sustainability Institute and stakeholders create projects to find solutions for the issues surrounding climate change by using renewable energy sources and creating innovative ideas to use energy. One of those projects is the IRIS project. Among other things, the IRIS project focuses on the renovation of 12 social housing buildings in the neighbourhood Kanaleneiland to near energy neutral buildings. The first support measurement in the first building to get 70 % of the residents to support the renovation has failed. This means that the renovation could not take place, and goals of the project can not be met. Therefore, a new method to increase the participation and engagement of the residents of Kanaleneiland was needed. The past several months I have focussed on creating, executing and evaluating a pilot with a new method named the 'Personal Approach Pilot', from an applied cognitive psychological perspective. Moreover, an important part of the pilot was the installation of a smart thermostat 'Toon' from Eneco. The Toon served as a playful and interactive tool to introduce the residents to the subject of sustainability, by giving them an opportunity to easily

get more insight in their energy consumption. To evaluate if this tool is effective within this population of Kanaleneiland with many second and third generation migrants and people of lower socio-economic status, I also had the task to execute a small usability study. The deliverable to Utrecht Sustainability Institute will be an advisory report with an evaluation of the pilot and the results of my usability research.

Location project (name company or University): Utrecht Sustainability Institute

External supervisor: Arno Peekel

UU Supervisor: Dennis Schutter

Name Student: Anna-Sophie Laan

Size of project: 7,5 ECTS

Title project: Analysis of www.beeldvandezorg.nl

Abstract: The healthcare in the Netherlands is a lot in the news lately due to many different issues, but most of all because of the rising working pressure. Because almost all of the attention given to healthcare is focussed on the negative sides of this sector, the VvAA wanted to make a website to highlight the positive aspects of working in healthcare and let visitors show the beauty of these professions. In cooperation with a photographer they mapped a day in the working lives of 8 different healthcare practitioners, where you can see the beauty but also the struggles of their work. Furthermore, the website is a platform where people can start a dialogue about the problems in healthcare and maybe come up with solutions. Commissioned by the VvAA, Onstuimig made the website www.beeldvandezorg.nl My assignment is to analyze the website and write an advice report for the VvAA and Onstuimig according to the results. The advice report is mostly focussed on attracting more visitors to their website so more people will read what the VvAA is trying to carry out. Furthermore, the advice report will contain some advice on the structure of the website, so that Onstuimig can optimize the website itself.

Location project (name company or University): Onstuimig

External supervisor: Laurens van Klaveren

UU Supervisor: Larissa Meijer

Name Student: Julia van der Donk

Size of project: 12,5 ECTS

Title project: Customer Experience at RealConnections

Abstract: RealConnections is a B2B company that implements and supports cloud software such as Zendesk, Microsoft, Okta and Voys. Their product is a service, named Remote Expertise and RC implemented a buddy system in which an employee is linked to a customer's contact person. In order to improve, an explorative customer experience research is conducted by analyzing internal data, collecting data via a questionnaire from 63 customers and by in-depth interviews with 6 customers. Research found, with an NPS of 32 and high scores on items as; responsiveness, repurchase intentions, expectations, added value, knowledge and general satisfaction, RC to be one of the top performers of the Netherlands. From the results, interviews and comments customers made the research found that RC can improve by creating visibility both internally but also between customers, managing expectations and having clear communications.

Location project (name company or University): RealConnections

External supervisor: Erik Vos

UU supervisor: Larissa Meijer

Name Student: Pim Verhagen

Size of project: 27,5 ECTS

Title project: Developing an experimental setup for examining embodiment in a teleoperation setting

Abstract: [...]

Location project (name company or University): TNO Soesterberg (Perceptual and Cognitive Systems)

External supervisor: Irene Kuling

UU Supervisor: Krista Overvliet

Name Student: Claire Borst

Size of project: 12.5 ECTS

Title project: Could the buying preference for magazines be predicted by implicit associations on affect and intention to buy?

Abstract: According to the Theory of Planned Behaviour consumer attitudes affect consumer behaviour intentions (Ajzen, 1991). This theory is taken into account to investigate whether an Evaluative Priming Test is effective to predict buying preferences by emotional responses and reasons to buy. This is done by implicit measures, because the acquired impulsive reactions are of better influence in driving behaviour than explicit measures (Fazio et al., 1986). Current study used implicit reaction time scores from two Evaluative Priming Tasks, which is used to assess evaluative responses to any object that is presented as a prime-stimulus in the priming test (Gawronski, 2014). The Forced Choice Priming Task is a speeded judgement task in which participants have to categorize targets in the corresponding category after being primed with several magazine covers. In the Free Choice Priming task participants were free to categorize different magazine covers into the categories buying or not buying. Results show that stimuli based on emotional responses (Affect) together with stimuli based on reasons to buy (Intention to buy) are coherent in predicting stimuli of buying preference. Additionally, participants reacted on average slower on negative stimuli than positive stimuli. These findings suggest that participants experienced on average weaker associations with negative stimuli from Affect and from Intention in combination with all magazine covers.

Location project (name company or University): Bloakes B.V.

External supervisor: Guus van Loon

UU Supervisor: Nico Romeijn

Name: Helen Visser

Size of project: 12 ECTS

Title project: Aircrafts and the quest for respite

Abstract: Generally, (aircraft) sounds are considered to have negative effects on (mental) health. Residents who live nearby airports complain about too little periods of rest between flights. Literature research has been conducted to collect methods for exploring the effects of temporal distribution of respite during aircraft overflight and it's effects on noise annoyance. The aim of this study is to find a applicable method for conducting a pilot experiment in a simulated environment to explore the effect of respite on noise annoyance.

Location project: Netherlands Aerospace Centre

External supervisor: Jaap Beintema

UU supervisor: Nico Romeijn

Name Student: Bianca Beerepoot

Size of project: 27,5 ECTS

Title project: "Hey Google, which smart speaker should I use?"

Abstract: This research for Valsplat focused on gathering more insight into the usage of voice assistants to be able to give better advice to clients. The main research question that has been answered with this research is: "When will a multimodal interface provide a better user experience than a unimodal interface for a specific use case?". The literature study, focused on characteristics of voice, audio, visuals and cognitive theories, showed that there are various use case characteristics

that should play a decisive role in developing for a uni- or multimodal interface. Based on the literature study a classification has been made, consisting of several use case characteristics that should influence the decision on which type of interface to use and several contextual factors that should also be taken into account. The classification can be used as an informative tool for the decision-maker. An experiment has been conducted with 85 participants that interacted with either a uni- or multimodal interface. The participants conducted two tasks, after which they assessed their satisfaction level using a questionnaire consisting of four questions about task completion, time, effort and subjective level of satisfaction. The experiment only tested the complexity characteristic of the classification. A higher level of satisfaction was expected for the more complex task when interacting multimodally. The result showed no significant difference between the two smart speakers. This means that user satisfaction did not increase when the participants interacted multimodally and the task became more complex. It can be concluded that there is a large grey area when choosing for the most appropriate smart speaker since it is not black-and-white when a uni- or multimodal interface should be used. Experimenting with the specific use case at hand might give the best insights.

Location project (name company or University): Valsplat

External supervisor: Stijn Nieuwendijk

UU Supervisor: Nico Romeijn

Abstracts poster presentations

Name: Molly Adams

Size of Project: 12.5 ECTS

Title of project: The Influence of the Big Five on Driving Style During Manual and Automated Driving and the handover of Control: A Preliminary Analysis of Participant Data

Abstract: Automated driving systems (ADS) are currently being tested all over the world and their presence on public roads is no longer a distant concept. With predicted benefits such as sustainability, accessibility and safety, it is likely that the widespread presence of these vehicles will be beneficial to society. However, there will still be error-prone humans at the end of these systems. Research shows that there exists a great deal of variation in driving styles, dependent on a range of characteristics and traits. In manual vehicles, personality has been shown to predict both safe and risky driving behaviours. However, we do not yet know how personality will influence behaviour in ADS and this is the topic of investigation of this research project. So far, demographic and personality information have been gathered from participants in preparation for driving simulations that will be conducted to assess driving style. The current research addresses the analysis of the participant data gathered so far. Analyses reveal possible evidence for a 'tech lover' stereotype, with older, male participants having more experience with ADS than younger and female participants across most personality dimensions. With the exception, however, of 'agreeable' females who have more experience in ADS than agreeable males. These initial analyses also reveal another personality exception. Analyses show that having more education generally predicts less driving experience, with the exception of the most open-minded participants.

Location of project: TU Delft

External supervisor: Daniël Heikoop

UU Supervisor: Joke Baas

Name Student: Josefine Funnekotter

Size of project: Internship = 12,5 ECTS

Title project: Change Management at AkzoNobel

Abstract: In AkzoNobel, I analysed the effectiveness of the internal communication approach in several different projects. I used a mixture of theories, tools and methods to find out how to improve

the effectiveness and evoke behavioural change. The results so far have lead to a Change Plan; an advisory/strategical document.

Location project (name company or University): AkzoNobel

External supervisor: Anne Wilhelmus

UU Supervisor: Gabriël Beckers

Name: Iga Jarosz

Size of Project: 12,5 ECTS

Title of project: An Experimental Manual to Python: The Stroop Task from Scratch

Abstract: Since my report is a manual, here I insert the preface rather than an abstract. Dear reader, In this manual you will find instructions to build, test and analyse a renown cognitive experiment called the Stroop task. The Stroop effect shows when we experience cognitive interference, for example, reading the word RED written in green font slows down the reaction time of naming the colour of the word (rather than the meaning of the word) in comparison to reading the word RED written in red font. Many variants of the original paradigm have been developed to study different aspects of this phenomenon. In case you haven't heard of the Stroop effect yet, don't worry you can have a closer look at it in the first section of the manual.

The experimental cycle will be described on a beginner and a more advanced level. This will be done by dividing the instructions into two parts. You will get to decide whether using a straight-forward interface of PsychoPy (beginner) or writing your own code (more advanced) is a more attractive alternative to construct your experiment. I will provide additional comments and insights concerning both approaches for a more informed decision. Using the example of the Stroop task, this manual will walk you through a part of what Python can offer for cognitive research. The intention is to give you a clear picture of when and how Python can help you make things faster and better in your study. Hence, at least parts of this manual should come in handy when studying other cognitive phenomena.

Together with the step-by-step guidelines you will be given a set of resources that I used to learn the basics of Python. As a user of manuals similar to this one, I don't want to bother you with extensive explanations concerning the mere installation of the programming environment or how simple operations can be executed. There already are many great resources to learn Python online, so if you need extra training, this manual will provide many suggestions. When I started creating this manual, I was new to programming myself. I intend to use it as an asset because I can provide information from a newbie perspective. Knowing that, try to see this manual as a set of notes that I have collected while learning on my own, so that you don't have to get stuck at the same points. Enjoy the struggle!

Location of project: Utrecht University

External supervisor: no external supervisor

UU Supervisor: Yentl de Kloe

Name student: Julia Meijer

Size of project: 12,5 ECTS

Title Project: Reviewing of the current jam-tail support system and a future in-vehicle jam tail support system: A literature study on human factors

Abstract: The driving task is changing, because of more information in vehicle. Drivers must process and comprehend a lot more information than in the past. Developments on giving jam tail support in the car via in-vehicle support systems (IVIS) are taking place. Certain aspects on human factors need to be reviewed to properly give this information. Some of these aspects are; modality, amount of information and personalization. These aspects can have benefits if used correctly.

Location project: Rijkswaterstaat

External Supervisor: Boris van Waterschoot

UU Supervisor: Stella Donker

Name student: Nina Geldof

[...]

Name Student: Nicole Plungjan

Size of project: 12.5 ECTS

Title project: Evaluating the effect of spectral composition of LED lighting on pleasantness in office environments

Abstract: With the introduction of connected LED lighting, office lighting can be tuned to the benefit of office workers more than ever. Some light spectra that are rich in the blue part, for example, have proven biological benefits including increased alertness and improved circadian rhythm. However, biologically effective light is not always the most pleasant to experience for people. We need to investigate how different light spectra with positive biological effects are appreciated by office workers to find an optimal balance between biological impact and appreciation/pleasantness.

Location project (name company or University): Signify

External supervisor: Bernt Meerbeek & Dzmitry Aliakseyeu

UU Supervisor: Krista Overvliet

Name student: Prithvi Dheeraj Cheraku Mulanti

Size of project: 12.5 ECTS

Title project: Understanding trends in Virtual Reality (VR) training and development of a VR Training module.

Abstract: In recent years there have been many advancements in the field of Virtual Reality and it is making its way into different fields. One such application of virtual reality is for training the employees. Many big companies like Walmart and Audi have already started employing VR for employee onboarding and training. I am interning at a Startup called BehaVr Academy which is making training modules for employees. One of my task was to look into which companies were using VR for training and how they were using it with special focus on the hospitality industry. I have compiled the findings from different research studies and delivered a report which includes the findings, benefits and how companies are employing VR for training. This information which I provided to the company is being used for their pitching. Then I started looking for information on training provided by the hotels to their employees. This helped me understand which training can be best delivered through VR medium. Using this information I worked on scenario building and script writing for training front desk employees in hotels. I am currently working on scenario building and script writing for sensitivity training and complaint management.

Location project: BehaVR Academy, Amsterdam

External Supervisor: Milan Collin

UU Supervisor: Anouk Keizer

Name Student: Lisa Verheul

Size of project: 12.5 ECTS

Title project: The experience of military: Interacting with a mixed-reality platform and with new technologies.

Abstract: Can the mixed- reality platform be used as an alternative for a field study to measure cognitive workload of military and to measure the influence of technologies on cognitive workload. In this study three technologies are being used: tactical display, tactile belt and augmented reality.

Location project (name company or University): TNO

External supervisor: Maurice van Beurden

UU Supervisor: Josien Meessen

Name Student: Job Davidse

Size of project: 12.5 ECTS

Title project: Designing web pages: text, picture and color use

Abstract: I did my internship at [Vergelijkdirect.com](https://vergelijkdirect.com). This company has a website on which visitors can compare and buy insurance policies. [Vergelijkdirect.com](https://vergelijkdirect.com) wants to sell as many insurance policies as possible. I researched the way in which page elements (text, picture, color use) influence visitor behavior towards buying insurance. The research was done using Google Optimize and Hotjar. Google Optimize was used to conduct AB tests. Hotjar was used to analyze the behaviour of page visitors. Ultimately, no significant differences emerged from the AB tests. Except for one outlier, whose reliability had yet to be confirmed. That there appeared to be no significant differences could be due to too small sample sizes. The Hotjar analysis showed that the mobile versions of two web pages were better at directing visitors towards buying insurance than the desktop version of those pages. This could be due to the fact that there were more irrelevant stimuli present on the desktop version of the two pages.

Location project (name company or University): [Vergelijkdirect.com](https://vergelijkdirect.com)

External supervisor: Bart Houtman

UU Supervisor: Larissa Meijer

Name Student: Lars Dijkstra

Size of project: 7,5 ECTS

Title project: Does your commercial make the most bang for its buck?

Abstract: Traditional commercial effectivity studies lack objectivity. How can we leverage advanced neuro techniques like eye tracking and EEG data to gain more insight into the effect of commercial positioning and commercial effectiveness.

Location project (name company or University): Unravel research

External supervisor: Tom van Bommel

UU Supervisor: Martijn Mulder.

Name Student: Vani Pillay

Size of project: 12.5 ECTS

Title project: Changing Minds to Change the Future

Abstract: Climate change is undeniable and a big threat to life as we know it. A gap between being concerned about climate change and acting on it has become increasingly apparent. It is widely believed that this gap is strengthened by various psychological barriers. Havas Lemz wants to gain further understanding of this gap to improve the performance of the Regreen Revolution campaign. Building on the work of Gifford (2011) and others, this paper investigates the strength of the 7 psychological barrier categories on the Dutch population. General barriers were investigated using an online survey (n=98), and individual interviews (n=7) were used to investigate the public's opinions on the campaign. *Social Comparison with Others* was identified as the strongest barrier, and *Discredence* as the weakest. Trends detected during the interviews included impact information, positive orientation, involvement of locals and a simple solution. Recommendations for further campaigns were provided, and a new potential campaign was proposed.

Keywords: Climate change, global warming, pro-environmental behaviour, psychological barriers, communication

Location project (name company or University): Havas Lemz

External supervisor: Tim Claassen

UU Supervisor: Krista Overvliet

Name Student: Elina Heeres

Size of project: 12,5 ECTS

Title project: The comparison of icons and words in pricing tests

Abstract: Consumer behavior is majorly influenced by unconscious mental processes. Neuromarketing uses implicit research techniques in order for companies to discover what consumers like or dislike. Neuropricing is one of those techniques. A neuropricing task measures response times to indicate the implicit willingness to pay for a product or service. This study was created to explore the improvements that could be made in terms of presentation. Currently, the pricing test of Bloakes provides textual information next to the product, but this could distract the participant from the product. This study explores whether icons could be an alternative or addition to text. Icons are symbolic signs designed to associate a picture with a meaning and are seen as efficient communicators. This study combined an eye tracker with the implicit pricing test and a total of twenty-three individuals participated. This study produced a pricing test with a text block and an icon block, where the feature information is either textual or presented through icons. There were two conditions, the text-icon and the icon-text condition. The icon and text blocks were compared in Total Visit Duration, Time to First Fixation, response time and price judgement. Results demonstrated that the Total Visit Duration on the products was longer in the icon block than in the text block. This means that participants looked longer at the products if products were paired with icons. Participants also looked longer at the text when compared to icons. The Time to First Fixation was shorter for text than for icons. The icons or text did not influence the price judgement, but the response times were longer during the icon block, which might suggest unfamiliarity. Based on these results, the study concluded that icons could be implemented in pricing tests, provided that the icons are familiar and simple. However, caution is advised, because icons can be more ambiguous than words as indicated by the response times.

Location project : Bloakes

External supervisor: Guus van Loon

UU supervisor: Nico Romeijn

Name Student: Ditte van Haalen

Size of project: 12,5 ECTS

Title project: Nudging safety behaviour

Abstract: One of the main goals in the field of human factors is to minimise erroneous decisions in safety-critical situations. Traditionally, the approach within the field is to take away the risk by making it (nearly) impossible for an error to happen and to mainly focus on conscious behaviour of human beings. However, such an approach might not be enough since human error often results from the fact that humans do not always make rational choices, but are influenced by *irrelevant* features in their decision making processes (Hansen & Jespersen, 2013). These irrational choices are often based on unconscious psychological processes, such as the use of biases and heuristics when facing decisions. A new approach to these kind of problems that could be an addition to safety management systems, is the use of **nudges**: a gentle push in the right direction. A nudge can be an implicit strategy to target these irrational factors in decision making processes. It is however important that nudges are not arbitrarily implemented, but have been carefully selected and evaluated. Knowledge of underlying psychological mechanisms, as well as the different nudges is therefore necessary. This report provides an overview of some of the most important psychological mechanisms in safety situations, 10 nudges that can be used in such situations and finally proposes a 6-step model based on literature that can be followed when considering implementing a nudge in safety-critical situations. Lastly, the model is demonstrated on the basis of a current case of Intergo.

Location project (name company or University): Intergo

External supervisor: Richard van der Weide

UU Supervisor: David Terburg

Name Student: Yongrui Zhao

Size of project: combine ECTS

Title project: Determining how motivated and interested job applicants are by Machine Learning

Abstract: People are one of the most important assets of companies, but Hiring employees is one of the largest and most costly problems for organizations (Johari, Yean, Adnan, Yahya, & Ahmad, 2012), The interview is one of the widely used technology to assess people's ability and personality, but it is also a subjective and problematic process. We all know that human beings are prone to bias, error and thus their decision making can always be inaccurate (Yeung, N., & Summerfield, C., 2012). As for interview, the most relevant causes of bias are gender and race (Salazar, M. K., 1990). Assuming adequate data and computing resources and if a strong theoretical understanding of the problem is available then a full numerical model is perhaps the most desirable solution. So far, they were successful in developing models for applicants' general motivation, perceived stress levels, and nervousness, in which the physiological signals explained between 16% and 18% of the variance. However, the explained variance showed is not good enough for companies' recruiters since it is only strong in distinguishing between groups of motivated and non-motivated applicants in general but for a company, they only want to interview the top 5 motivated applicants. To expand current results, advanced methods and machine learning models should be applied, such as principal component analysis, artificial neural network, support vector machine and cluster. Those machine learning models are more capable of non-linear regression and thus can bring forth models with better predictive power.

Location project (name company or University):Neurolytics

External supervisor: Felix Hermsen

UU Supervisor: Marnix Naber

Name student: Anaïs Thijssen

Size of project: 7.5 ECTS

Title project: Determining genetic variations that could influence fear conditioning

Abstract: Previous work identified differences in fear conditioning trajectories that could have important implications in a clinical setting. The goal of my internship was to find genetic variations that could explain these individual differences. To find possible genetic variations if interest, a selection procedure was developed to go from a general trait (fear extinction) to specific mutations that could have an influence on this trait.

Location project: Utrecht University

UU supervisor: Joke Baas